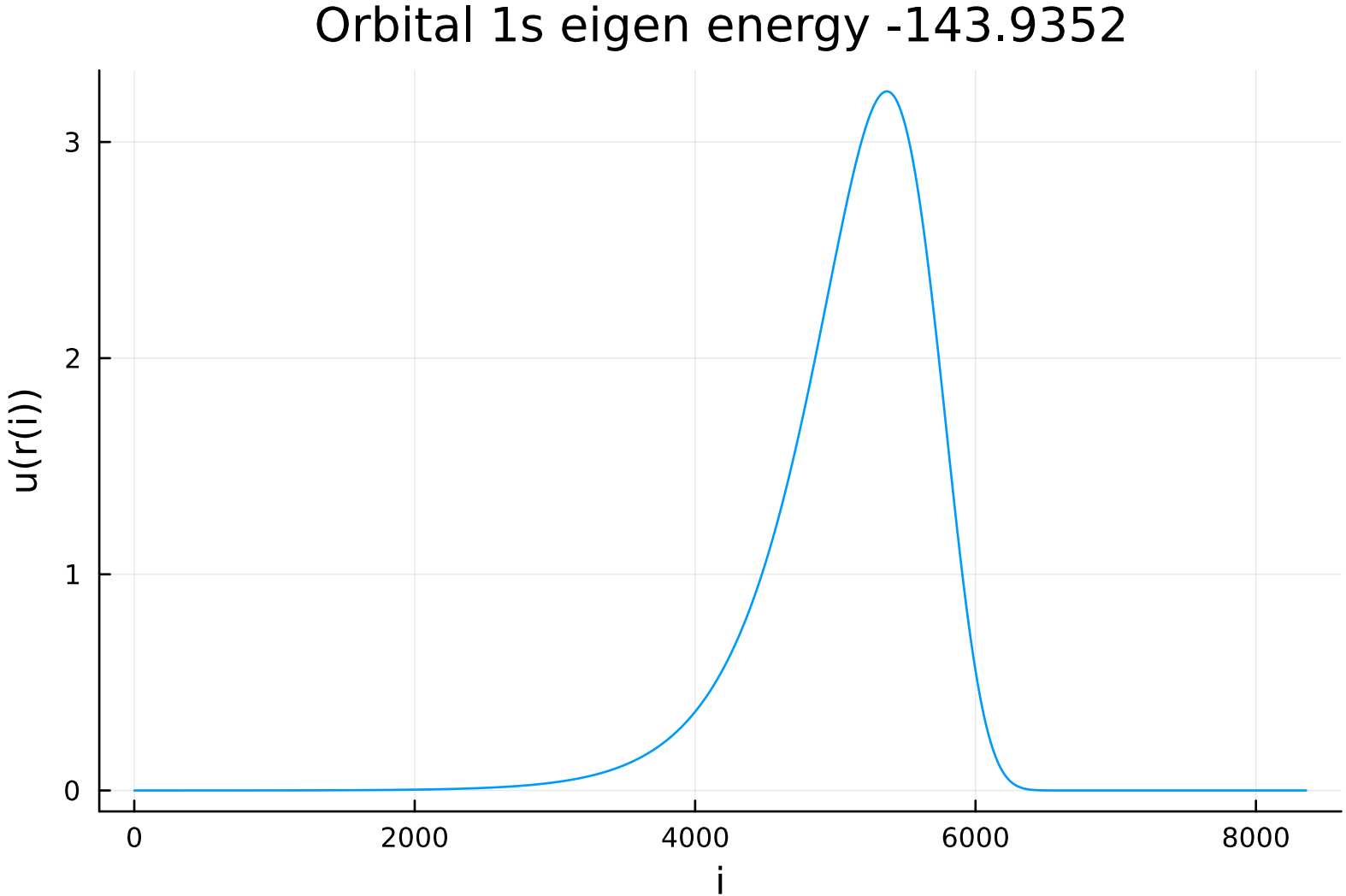
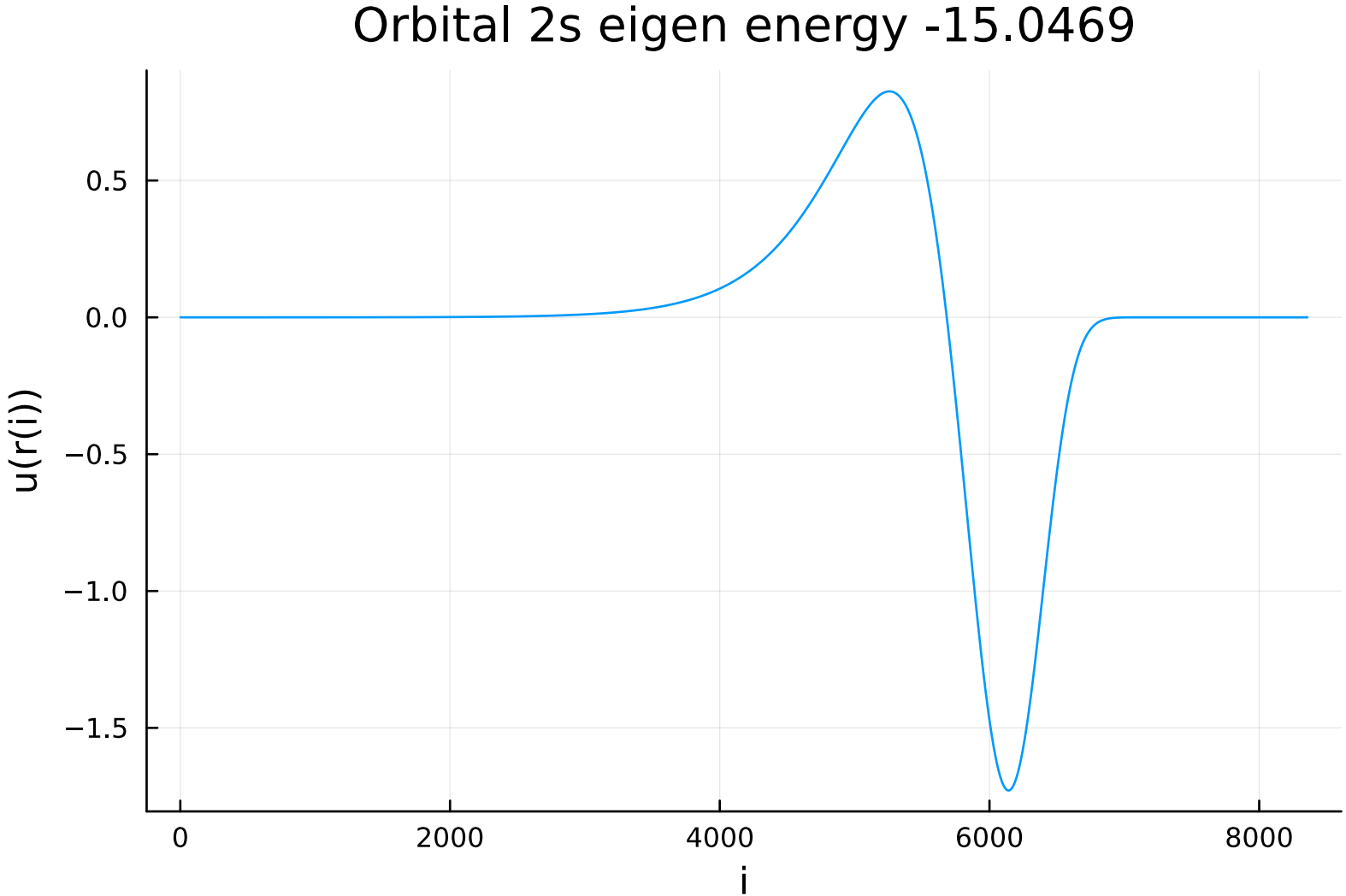
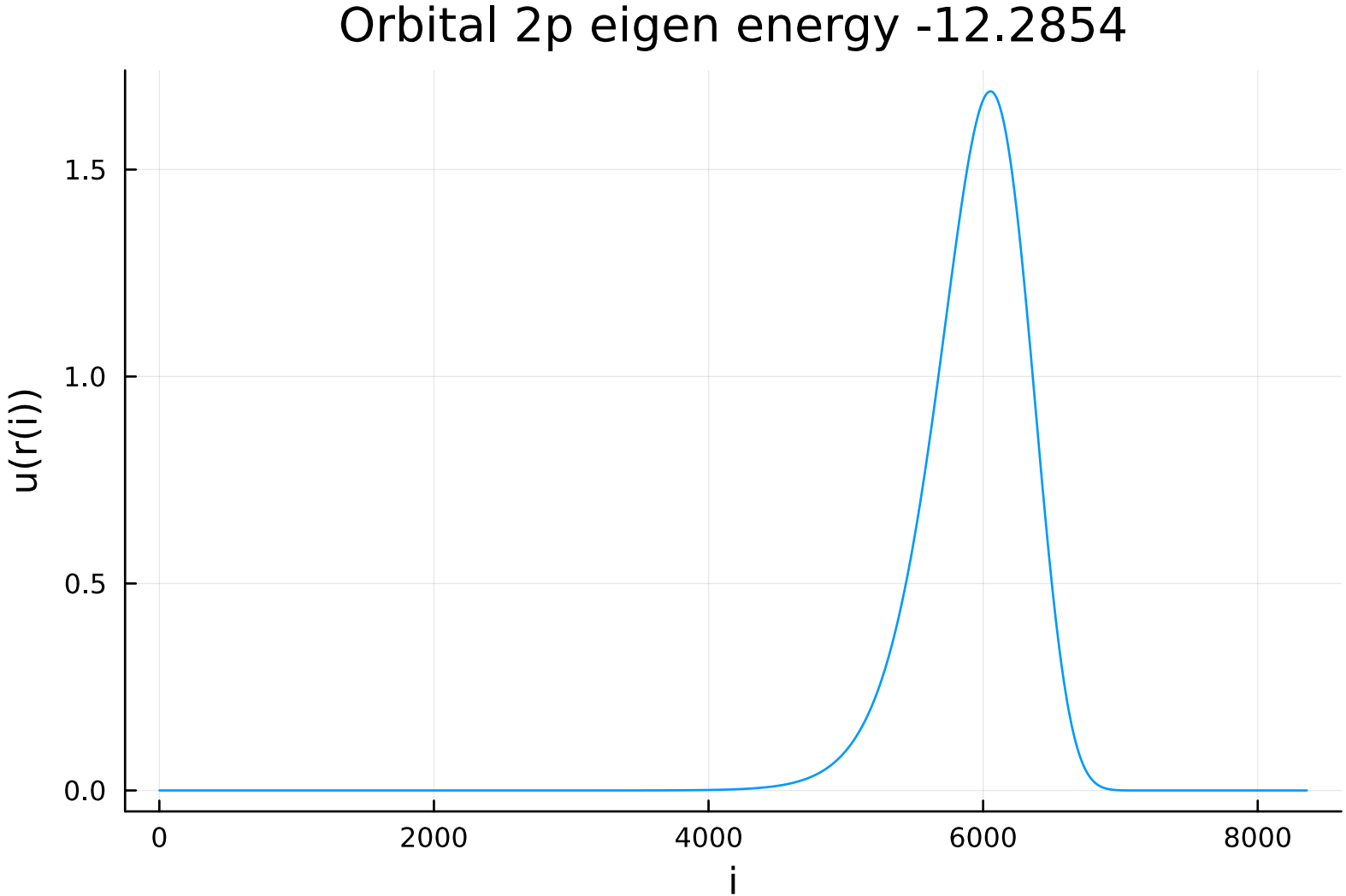
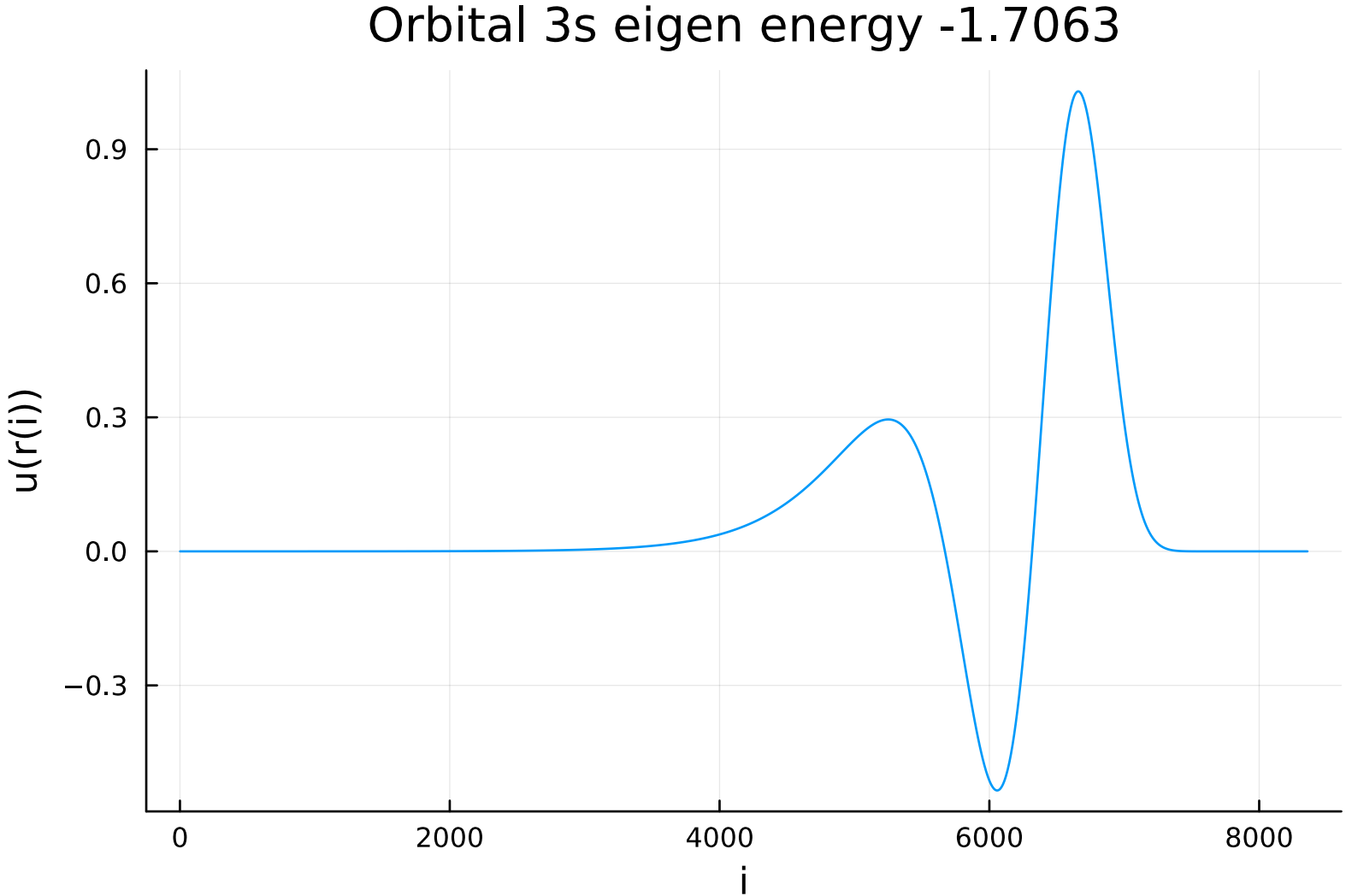
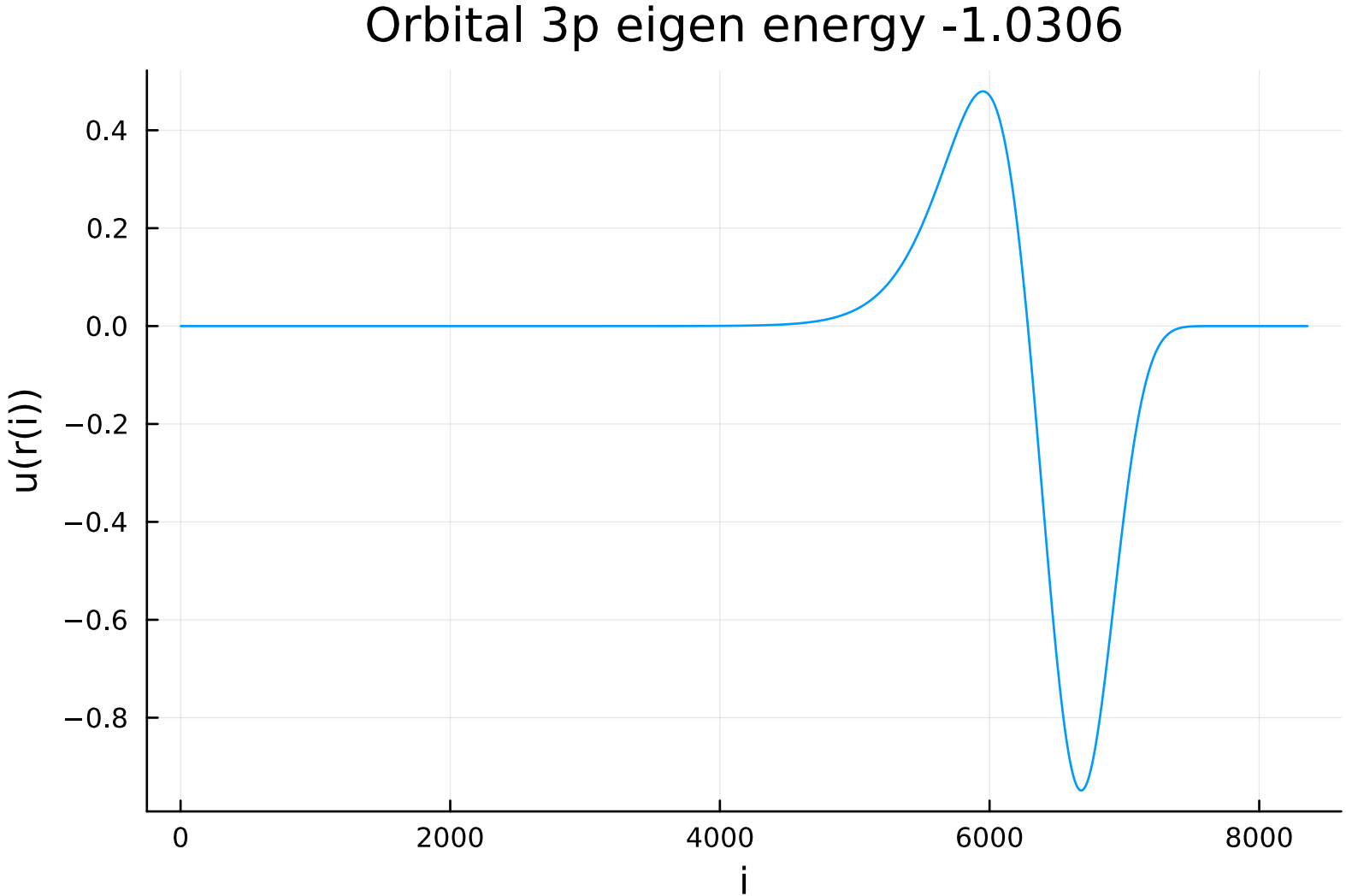
Calcium	E in Hartree			
Name	E NIST	E pred	|ΔE|	|ΔE|/E NIST
Ecoul	285.206130	285.206125	0.000005	0.000000
3p	-1.030572	-1.030573	0.000001	-0.000001
4s	-0.141411	-0.141410	0.000001	-0.000004
Etot	-675.742283	-675.742283	0.000000	-0.000000
2s	-15.046905	-15.046906	0.000001	-0.000000
3s	-1.706331	-1.706331	0.000000	-0.000000
Exc	-34.142538	-34.142538	0.000000	-0.000000
Ekin	674.657334	674.657333	0.000001	0.000000
2p	-12.285376	-12.285376	0.000000	-0.000000
1s	-143.935181	-143.935181	0.000000	-0.000000

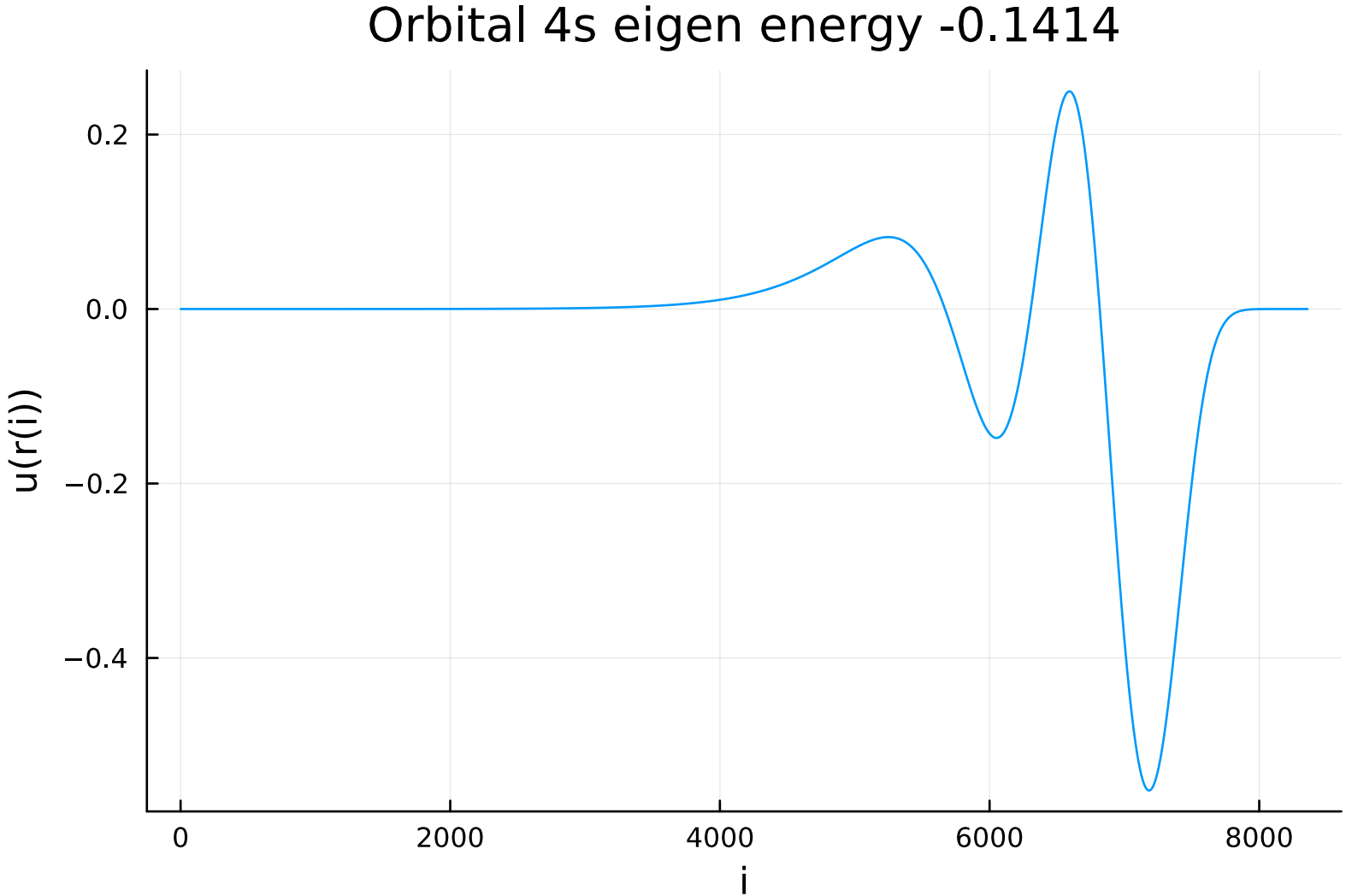

Orbital 1s eigen energy -143.9352
u(r(i))
0
1
2
3
0
2000
4000
6000
8000
i


Orbital 2s eigen energy -15.0469
0.5
0.0
−0.5
−1.0
−1.5
u(r(i))
0
2000
4000
6000
8000
i


Orbital 2p eigen energy -12.2854
u(r(i))
1.5
1.0
0.5
0.0
0
2000
4000
6000
8000
i


Orbital 3s eigen energy -1.7063
u(r(i))
0.9
0.6
0.3
0.0
−0.3
0
2000
4000
6000
8000
i


Orbital 3p eigen energy -1.0306
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
u(r(i))
0
2000
4000
6000
8000
i


Orbital 4s eigen energy -0.1414
0.2
0.0
−0.2
−0.4
u(r(i))
0
2000
4000
6000
8000
i